

- 3) Explain, with the help of an example, the concept of simultaneity bias.

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## 16.7 IDENTIFICATION CONDITIONS

There are two conditions that must be fulfilled for an equation to be identified.

### 16.7.1 Order Condition of Identification

This condition is based on a counting rule of the variables included and excluded from a particular equation. It is a necessary but not sufficient condition for identification of an equation. The order condition may be stated as follows:

For an equation to be identified the total number of variables (endogenous and exogenous) excluded from it must be equal to or greater than the number of endogenous variables in the model less one. Remember that in a simultaneous equation system, the number of endogenous variables is equal to the number of equations of the model. Thus, the order condition can be stated as follow:

For an equation to be identified the total number of variables excluded from it but included in other equations must be at least as large as the number of equations of the system less one.

Suppose,

$G$  = Total number of equations (=total number of endogenous variables)

$K$  = Total number of variables in the model (endogenous and predetermined)

$M$  = Total number of variables, endogenous and exogenous, included in a particular equation

Then the order condition for identification may be symbolically expressed as

$$(K-M) \geq (G-1)$$

$$[\text{Excluded variables}] \geq [\text{Total number of equations} - 1]$$

### Example 16.6

Suppose a simultaneous model contains 10 equations with 15 variables (10 endogenous and 5 exogenous).

(a) Let us assume, for the first equation we have

|          |          |          |
|----------|----------|----------|
| $G = 10$ | $K = 15$ | $M = 11$ |
|----------|----------|----------|

Recall that the **order condition** is given by

$$(K - M) \geq (G - 1)$$

In this case, we have

$$(15 - 11) < (10 - 1)$$

Thus, the order condition is not satisfied, and the first equation is under-identified.

(b) Let us assume, for the second equation we have

|          |          |         |
|----------|----------|---------|
| $G = 10$ | $K = 15$ | $M = 5$ |
|----------|----------|---------|

Order condition:

$$(K - M) \geq (G - 1)$$

$$(15 - 5) > (10 - 1)$$

Thus, the order condition is satisfied for the second equation.

The order condition for identification is necessary for a relation to be identified, but it is not sufficient. Therefore, order condition may be fulfilled in an equation and yet the relation may not be identified.

#### 16.7.2 Rank Condition of Identification

The rank condition states that: in a system of  $G$  equations an equation is identified if and only if it is possible to construct at least one non-zero determinant of order  $(G-1)$  from the coefficients of the variables excluded from that equation but contained in other equations of the model.

This condition is called the rank condition because it refers to the rank of the matrix of parameters of excluded variables. The rank of a matrix is the order of the largest non-zero determinant which can be formed from the matrix. In our case the relevant matrix is the sub-matrix of coefficients of the excluded variables. Hence, the rank condition may be also stated as follows: a sufficient condition for identification of a relationship is that the rank of the matrix of parameters of all the excluded variables (endogenous and predetermined) from that equation be equal to  $(G-1)$

The practical steps for tracing the identifiability of an equation of a structural model may be outlined as follows:

**Step1:** Write down the parameters of all the equations of the model in a separate table, noting that the parameter of a variable excluded from an equation is equal to zero.

### Example 16.7

Consider the following simultaneous equations model:

$$y_1 = 3y_2 + 2x_1 + x_2 + u_1$$

$$y_2 = y_3 + x_3 + u_2$$

$$y_3 = y_1 - y_2 - 2x_3 + u_3$$

where the  $y$ 's are endogenous variables and the  $x$ 's are predetermined variables.

The above model can be written in the following form:

|                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------|
| $-y_1 + 3y_2 + 0y_3 - 2x_1 + x_2 + 0x_3 + u_1 = 0$ $0y_1 + y_2 + y_3 - 0x_1 + 0x_2 + x_3 + u_2 = 0$ $y_1 + y_2 + y_3 - 0x_1 + 0x_2 + 2x_3 + u_3 = 0$ |
|------------------------------------------------------------------------------------------------------------------------------------------------------|

Ignoring the random disturbances the table of the parameters of the model is as follows:

**Table 16.1: Parameters of the Equation System**

| Equations                | Variables |       |       |       |       |       |
|--------------------------|-----------|-------|-------|-------|-------|-------|
|                          | $y_1$     | $y_2$ | $y_3$ | $x_1$ | $x_2$ | $x_3$ |
| 1 <sup>st</sup> equation | -1        | 3     | 0     | -2    | 1     | 0     |
| 2 <sup>nd</sup> equation | 0         | -1    | 1     | 0     | 0     | 1     |
| 3 <sup>rd</sup> equation | 1         | -1    | -1    | 0     | 0     | -2    |

**Step2:** Eliminate the row of coefficients of the equation, which is being examined for identification.

For example, if we want to examine the identifiability of the second equation of the model, we strike out the second row of the table of coefficients.

**Step3:** Eliminate the columns in which a non-zero coefficient of the equation being examined appears. By deleting the relevant row and columns we are left with the coefficients of variables not included in the concerned equation but contained in other equations of the model.

For example, if we are examining for identification the second equation of the system, we will eliminate the second, third and the sixth columns of Table 16.1. Thus, after deleting the required row and columns, we obtain Table 16.2(b).

Table 16.2-(a),(b)

| Table of Structural Parameters |       |       |       |       |       |       | Table of parameters left<br>after eliminating the<br>relevant rows and columns |       |       |
|--------------------------------|-------|-------|-------|-------|-------|-------|--------------------------------------------------------------------------------|-------|-------|
|                                | $y_1$ | $y_2$ | $y_3$ | $x_1$ | $x_2$ | $x_3$ | $y_1$                                                                          | $x_1$ | $x_2$ |
| 1 <sup>st</sup>                | -1    | 3     | 0     | -2    | 1     | 0     | -1                                                                             | -2    | 1     |
| 2 <sup>nd</sup>                | 0     | -1    | 1     | 0     | 0     | 1     |                                                                                |       |       |
| 3 <sup>rd</sup>                | 1     | -1    | -1    | 0     | 0     | -2    | 1                                                                              | 0     | 0     |

**Step4:** Form the determinant(s) of order  $(G-1)$  and examine their value. If *atleast one* of these determinants is non-zero, the equation is identified. If all the determinants of order  $(G-1)$  are zero, the equation is under-identified.

From Table 16.2(b), we form determinants of order  $(G-1) = (3-1)=2$ . We obtain three such determinants:

$$\Delta_1 = \begin{vmatrix} -1 & -2 \\ 1 & 0 \end{vmatrix} \neq 0$$

$$\Delta_2 = \begin{vmatrix} -2 & 1 \\ 0 & 0 \end{vmatrix} = 0$$

$$\Delta_3 = \begin{vmatrix} -1 & 1 \\ 1 & 0 \end{vmatrix} \neq 0$$

(the symbol 'Δ' stands for determinant'). We see that we can form two non-zero determinants of order  $G-1 = 3-1 = 2$ ; hence the second equation of our system is identified.

**Step5:** To see whether the equation is exactly identified or over-identified we use the order condition  $(K - M) \geq (G - 1)$ . With this criterion, if the equality sign is satisfied, that is, if  $(K - M) = (G - 1)$ , the equation is exactly identified. If the inequality sign holds, that is, if  $(K - M) > (G - 1)$ , then the equation is over-identified.

In the case of the second equation, we have

$$G = 3 \quad K = 6 \quad M = 3$$

and the counting rule  $(K-M) \geq (G-1)$  gives

$$(6-3) > (3-1)$$

Therefore, the second equation of the model is over-identified. You should check the identification status of the first equation and the third equation. You will find that the first equation is exactly identified according to the order condition and the rank condition. Third, equation fulfils the order condition but does not fulfil the rank condition.

We illustrate the two formal conditions of identification with the help of an example.

### Example 16.8

Assume that we have a model describing the market of an agricultural product. From the theory of partial equilibrium, we know that the price in a market is determined by demand and supply. The main determinants of demand are the price of the commodity, the prices of other commodities, incomes, and tastes of consumers. Similarly, important determinants of supply are price of the commodity, other prices, technology, prices of factors of production, and weather condition. The equilibrium condition is that demand be equal to supply. The above theoretical information can be expressed in the form of the following mathematical model:

$$D = a_0 + a_1P_1 + a_2P_2 + a_3Y + a_4t + u$$

$$S = b_0 + b_1P_1 + b_2P_2 + b_3Y + b_4t + w$$

$$D = S$$

where

$D$  = quantity demanded

$S$  = quantity supplied

$P_1$  = price of the given commodity

$P_2$  = prices of other commodities

$Y$  = income

$C$  = costs (index of prices of factors of production)

$t$  = time trend. In the demand function it stands for 'taste'; in the supply function it stands for 'technology'

The above model is mathematically complete in the sense that it contains three equations in three endogenous variables:  $D$ ,  $S$  and  $P_1$ . The remaining variables,  $Y$ ,  $P_2$ ,  $C$ ,  $t$  are exogenous. Suppose we want to check the identification status of the supply function. We apply the two criteria for identification:

1. Order condition:  $(K-M) \geq (G-1)$

In our example we have

$$K = 7 \quad M = 5 \quad G = 3$$

Therefore,  $(K-M) = (G-1)$

$$(7-5) = (3-1) = 2$$

Consequently, the second equation satisfies the order condition for identification.

2. Rank condition: The table of the coefficients of the structural model is as follows:

**Table 16.3: Coefficients of the Structural Model**

| Equations                | Variables |       |       |       |       |     |       |
|--------------------------|-----------|-------|-------|-------|-------|-----|-------|
|                          | $D$       | $P_1$ | $P_2$ | $Y$   | $t$   | $S$ | $C$   |
| 1 <sup>st</sup> equation | -1        | $a_1$ | $a_2$ | $A_2$ | $a_4$ | 0   | 0     |
| 2 <sup>nd</sup> equation | 0         | $b_1$ | $b_2$ | 0     | $b_4$ | -1  | $b_3$ |
| 3 <sup>rd</sup> equation | -1        | 0     | 0     | 0     | 0     | 1   | 0     |

Following the procedure explained earlier, we eliminate the second row and the second, third, fifth, sixth and seventh columns. Thus, we are left with the table of the coefficients of excluded variables (see Table 16.4 (a) and (b)).

**Table 16.4-(a): Complete Table of Structural Parameters**

|                            |
|----------------------------|
| -1 $a_1a_2a_3a_4$ 00       |
| 0 $b_1b_2$ 0 $b_4$ 1 $b_3$ |
| -1000010                   |

**Table 16.4(b): After eliminating the relevant rows and columns**

|          |
|----------|
| -1 $a_3$ |
| -10      |

The value of the determinant is non-zero, provided that  $a_3 \neq 0$ .

$$\Delta = \begin{vmatrix} -1 & a_3 \\ -1 & 0 \end{vmatrix} = (0)(-1) - (-1)(a_3) = a_3$$

We see that both the order and rank conditions are satisfied. Hence, the second equation of the model is identified. Furthermore, we see that in the order condition the equality holds:  $(7 - 5) = (3 - 1) = 2$ . Consequently, the second structural equation is exactly identified.

Thus, we have seen that the order condition of identification is the necessary condition, and the rank condition is 'both necessary and sufficient condition' for identification of an equation. *It implies that even if we do not consider the order condition, we can find out the identification status of an equation through the rank condition.*

The order condition, however, is quite simple to apply. Therefore, we usually begin with the order condition. If the order condition is not fulfilled, we infer that the equation is under-identified. If the order condition is satisfied, we move on to the rank condition.

- 1) Consider the three-equation model

$$Y_1 = \beta_{13}Y_3 + \gamma_{12}X_2 + u_1$$

$$Y_2 = \beta_{21}Y_1 + \beta_{23}Y_3 + \gamma_{21}X_1 + \gamma_{22}X_2 + u_2$$

$$Y_3 = \gamma_{33}X_3 + u_3$$

where  $Y_1$ ,  $Y_2$  and  $Y_3$  are endogenous variables, and  $X_1$ ,  $X_2$  and  $X_3$  are exogenous variables. Find out the identification status of each equation of the model, based on the order condition and the rank condition.

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- 2) The structure of a model with four endogenous and three exogenous variables is as follows: (1 indicates the presence and 0 indicates the absence of a variable in the equation):

| $Y_1$ | $Y_2$ | $Y_3$ | $Y_4$ | $X_1$ | $X_2$ | $X_3$ |
|-------|-------|-------|-------|-------|-------|-------|
| 1     | 0     | 1     | 1     | 1     | 0     | 0     |
| 1     | 1     | 1     | 0     | 0     | 1     | 1     |
| 0     | 0     | 1     | 0     | 1     | 0     | 0     |
| 1     | 0     | 1     | 1     | 0     | 1     | 0     |

Which of the four equations is identified?

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## 16.8 LET US SUM UP

Unlike the single equation model, in simultaneous equations models, we get more than one dependent, or endogenous variable. This endogenous variable may appear in one equation as explained variable and, in another equation, as explanatory variable. Due to the presence of endogenous variables as explanatory variables in some of the equations, the regressors are stochastic in nature. In such situations the classical OLS regression method is not applicable as it may result in inconsistent estimators, and there is simultaneity bias.

Identification of an equation is necessary estimation of the equation. The objective of the identification problem is to obtain unique estimators of the structural parameters from the reduced form parameters. An equation in a

system of simultaneous equations is identified only when we can resolve this issue. If not, the equation remains in unidentified or under-identified situation.

An identified equation can be just-identified or over-identified. In the case of an exactly identified equation, unique estimators of structural parameters can be obtained. In the case of an over identified equation, on the other hand, more than one estimator for certain structural parameters is obtained.

There are two conditions for evaluation of identification status of an equation in a simultaneous equation system, viz., order condition, and rank condition. The order condition is a necessary condition for identification. On the other hand, the rank condition is a necessary and sufficient condition for identification. If the rank condition is satisfied, the order condition is satisfied too, although the reverse is not true.

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## 16.9KEY WORDS

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|                                    |                                                                                                                                                 |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Endogenous Variables</b>        | : These are variables which get determined jointly and interdependently within the model.                                                       |
| <b>Exactly-identified Equation</b> | : If there is sufficient information available to distinguish the equation then it is called just identified or exactly identified equation     |
| <b>Exogenous Variables</b>         | : These are variables which get determined outside the model, and independently of the endogenous variable.                                     |
| <b>Identification</b>              | : It refers to the possibility (or impossibility) of deducing the values of the parameters.                                                     |
| <b>Inconsistent Estimators</b>     | : The value of the estimates may not converge to the population parameter.                                                                      |
| <b>Over-identified Equation</b>    | : If there is more than sufficient information regarding the equation then the equation is called over-identified.                              |
| <b>Paradox of Identification</b>   | : It refers to the paradox that identification is done in terms of the variables absent from the equation not in terms of the present variable. |
| <b>Pre-determined Variables</b>    | : The set of exogenous variables and the lagged endogenous variables.                                                                           |
| <b>Reduced Form Models</b>         | : It refers to models where each equation contains only one dependent variable.                                                                 |
| <b>Reduced Form Parameters</b>     | : Parameters of the reduced form models.                                                                                                        |



|                                      |                                                                                                                                                                     |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Simultaneity Bias</b>             | : Bias resulting from applying OLS to the simultaneous equations model.                                                                                             |
| <b>Simultaneous Equations Models</b> | : Models where the equations are related and cannot be solved independently.                                                                                        |
| <b>Structural Equation</b>           | : Equations which represents the structure of an economic model or the behavior of an economic agent.                                                               |
| <b>Structural Parameters</b>         | : Parameters appearing in the structural equations.                                                                                                                 |
| <b>Under-identified Equation</b>     | : If the structure of the equation is not unique or there is not sufficient information to uniquely distinguish the equation then the equation is under-identified. |

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## 16.10 SOME USEFUL BOOKS

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Gujarati, D. N., D. C. Porter and S. Gunasekar, 2017, *Basic Econometrics*, Fifth Edition, McGraw Hill

Maddala, G. S., and K. Lahiri, 2012, *Introduction to Econometrics*, Fourth Edition, Wiley

Wooldridge, J. M., 2022, *Introductory Econometrics: A Modern Approach*, Seventh Edition, Cengage Learning

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## 16.11 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

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### Check Your Progress 1

- 1) Refer to Section 16.3 and answer.
- 2) Refer to Section 16.3 and answer. Exogenous variable is one of the categories of pre-determined variables.
- 3) The structure of a simultaneous equations model is given in Section 16.3. Refer to it and answer.

### Check Your Progress 2

- 1) Refer to Section 16.5 and answer.
- 2) In order to identify an equation, one has to examine the variables excluded from the equation.
- 3) You should show that OLS estimators are biased and inconsistent. Refer to Section 16.4.

### Check Your Progress 3

- 1) First apply the rank condition. Write down the equations as we have done in Example 16.7. Strike out the required row and columns. Check the rank condition.
- 2) Follow the same procedure as in Question 1.

